

Homework 2

1. a)

$$U_0(z) = X_0(z^2)$$

$$U_1(z) = X_1(z^2)$$

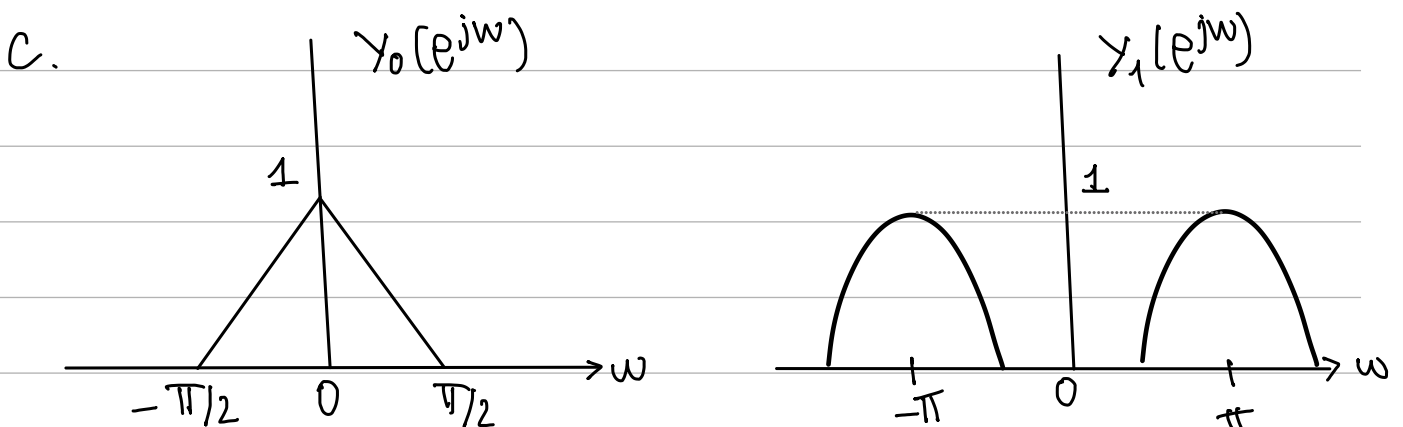
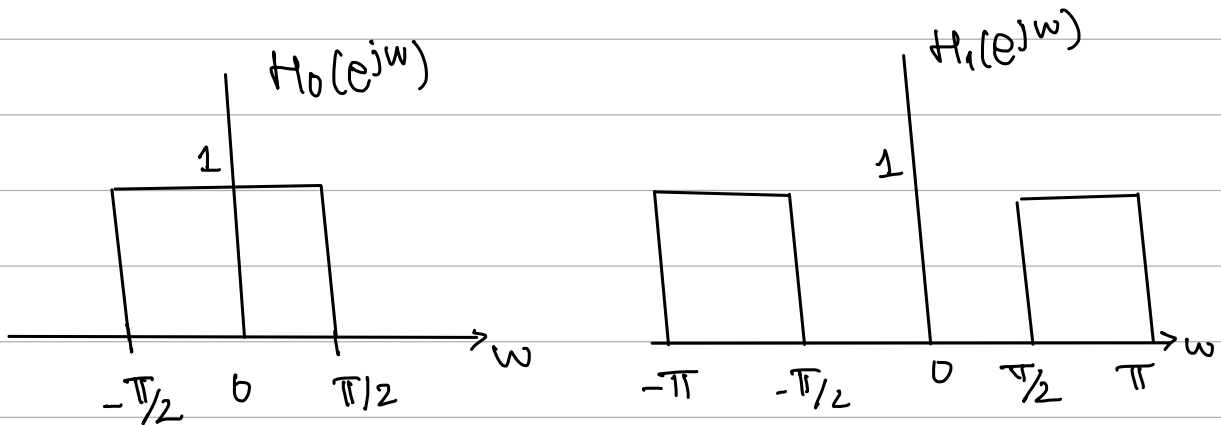
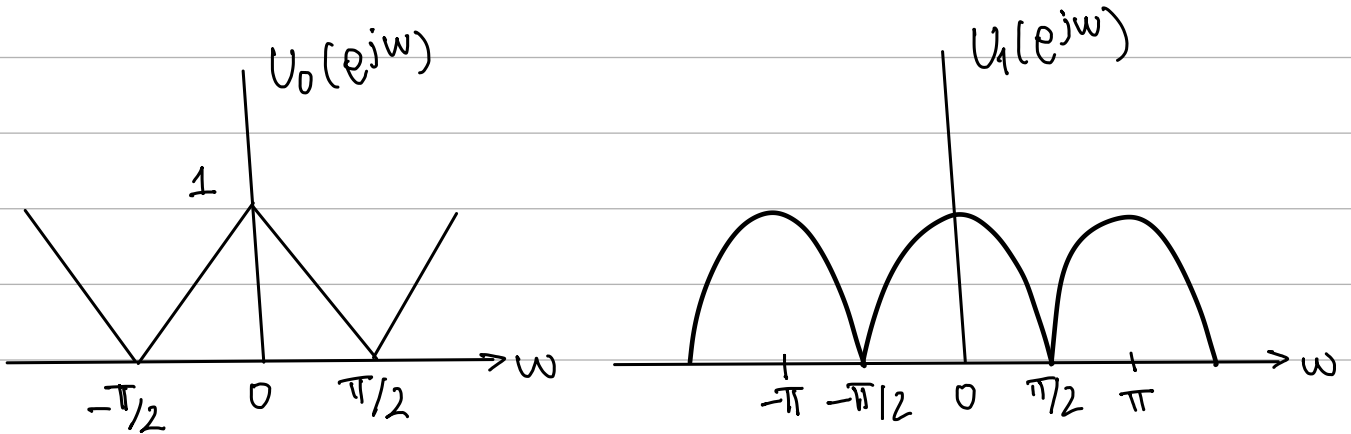
$$Y_0(z) = U_0(z) \cdot H_0(z) = X_0(z^2) \cdot H_0(z)$$

$$Y_1(z) = U_1(z) \cdot H_1(z) = X_1(z^2) \cdot H_1(z)$$

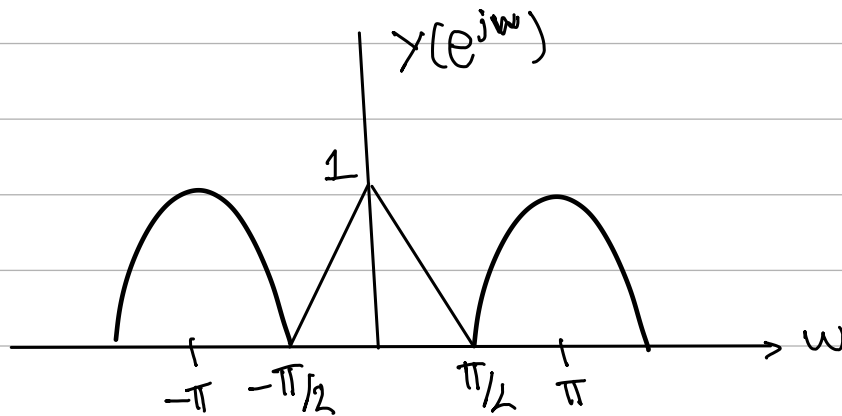
b)

$$U_0(e^{j\omega}) = X_0(e^{2j\omega})$$

$$U_1(e^{j\omega}) = X_1(e^{2j\omega})$$



$$y(e^{j\omega}) = y_0(e^{j\omega}) + x_1(e^{j\omega})$$



d. This is a transmultiplexer system. It interleaves two signal's frequency spectrum.

$$2. \quad a) \quad V_0(z) = X(z) H_0(z)$$

$$V_1(z) = X(z) H_1(z)$$

$$b) \quad U_0(z) = \frac{1}{2} [V_0(z^{1/2}) + V_0(-z^{1/2})]$$

$$= \frac{1}{2} [X(z^{1/2}) H_0(z^{1/2}) + X(-z^{1/2}) H_0(-z^{1/2})]$$

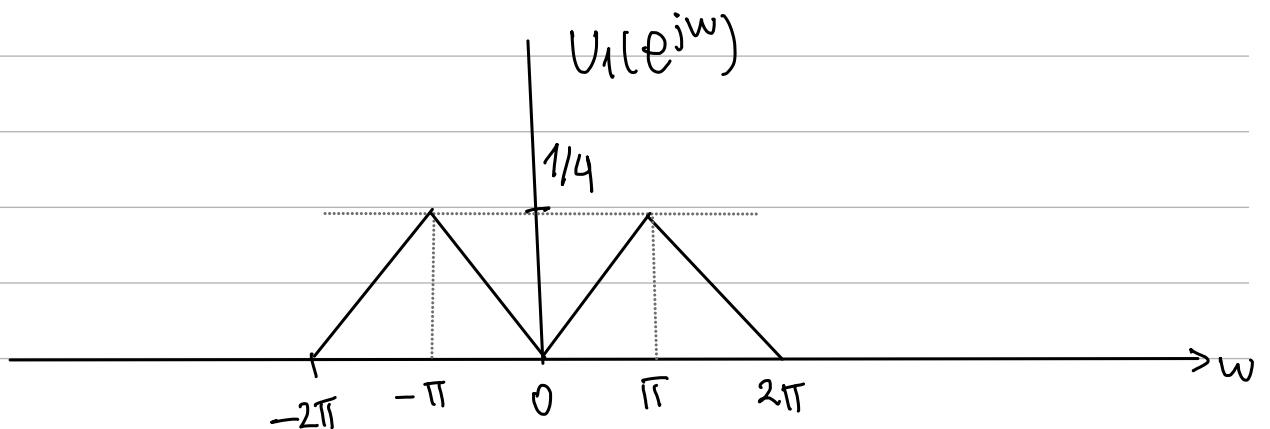
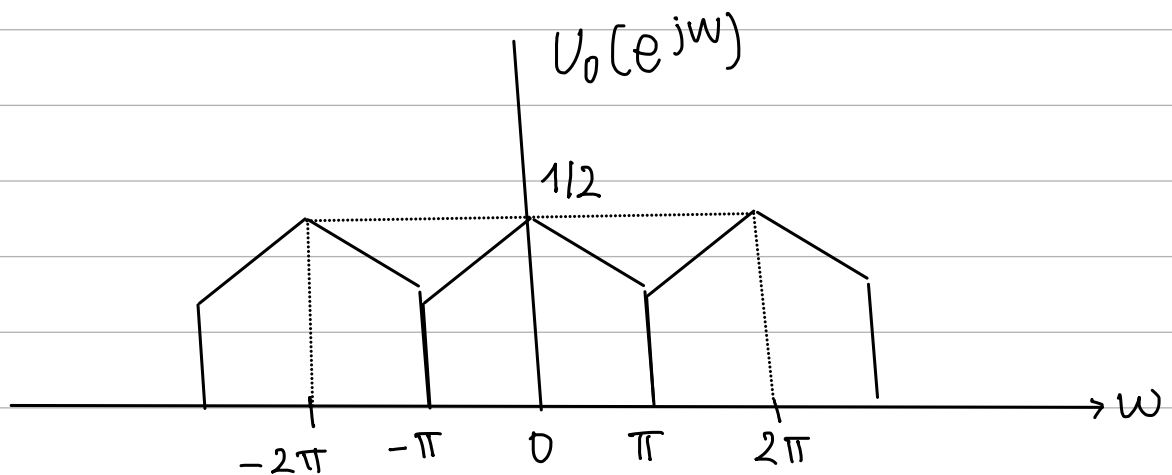
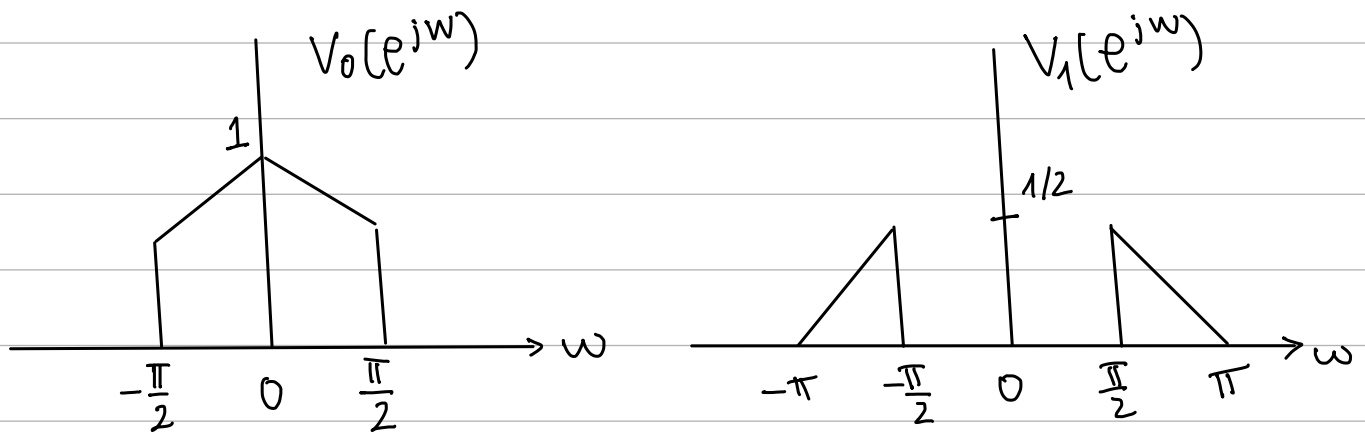
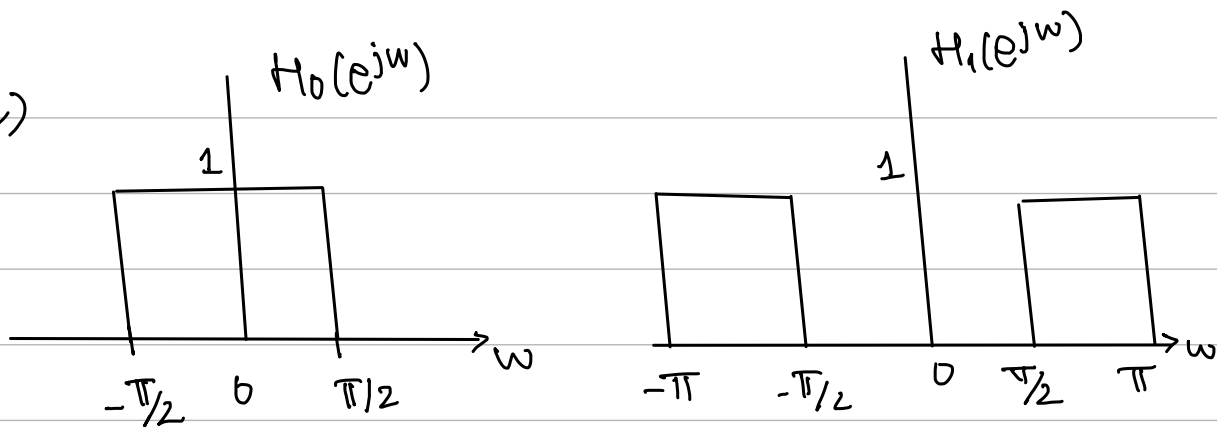
$$U_1(z) = \frac{1}{2} [V_1(z^{1/2}) + V_1(-z^{1/2})]$$

$$= \frac{1}{2} [X(z^{1/2}) H_1(z^{1/2}) + X(-z^{1/2}) H_1(-z^{1/2})]$$

$$U_0(e^{j\omega}) = \frac{1}{2} [V_0(e^{j\frac{\omega}{2}}) + V_0(e^{j(\frac{\omega}{2} - \pi)})]$$

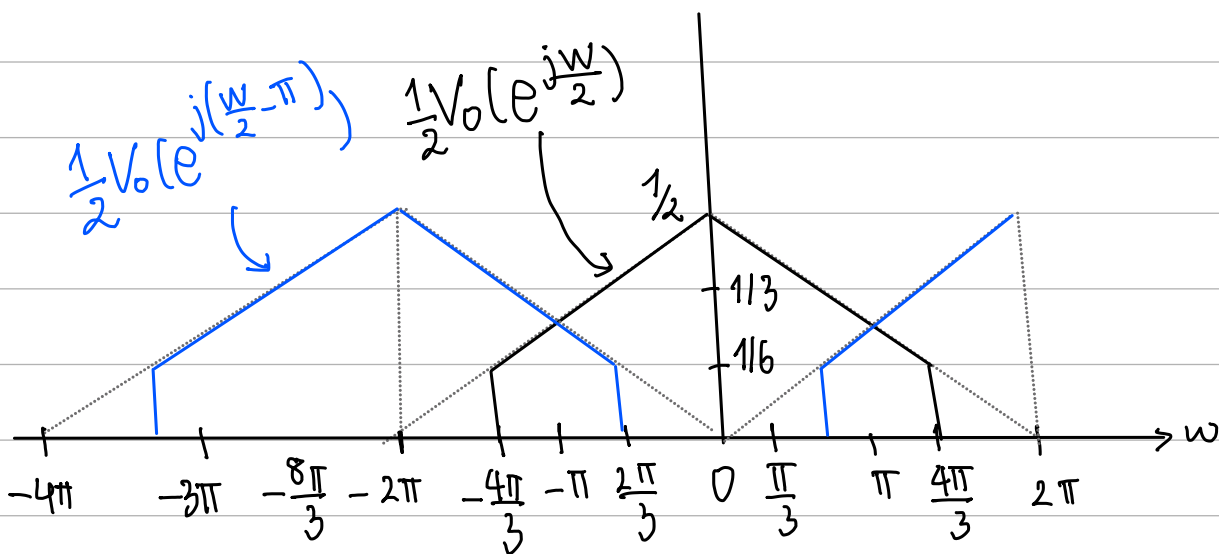
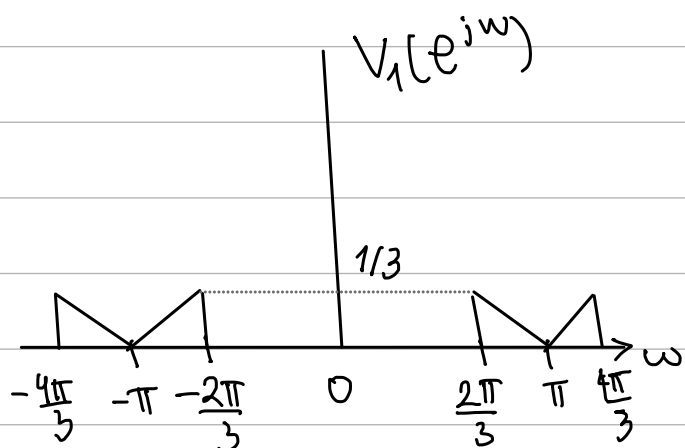
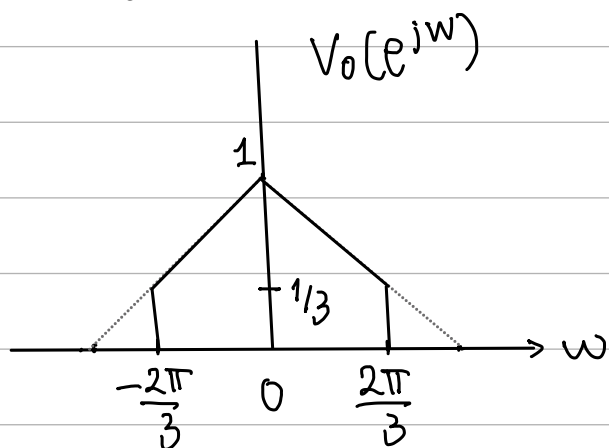
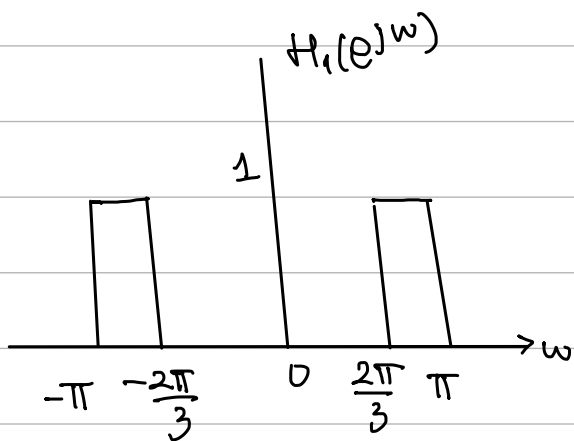
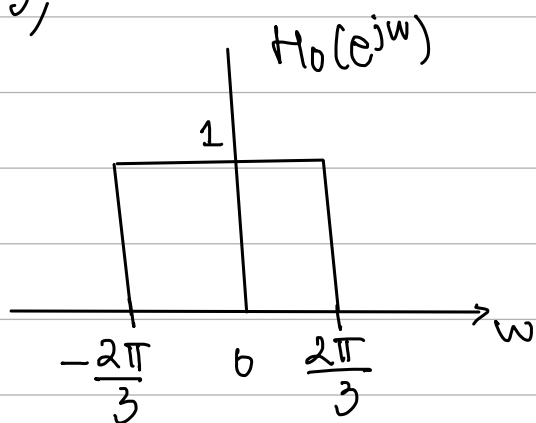
$$U_1(e^{j\omega}) = \frac{1}{2} [V_1(e^{j\frac{\omega}{2}}) + V_1(e^{j(\frac{\omega}{2} - \pi)})]$$

c)

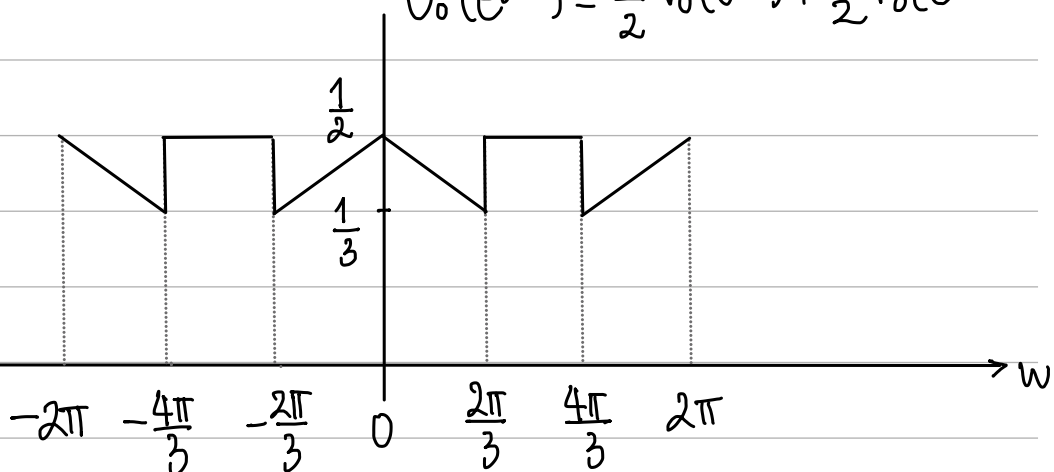


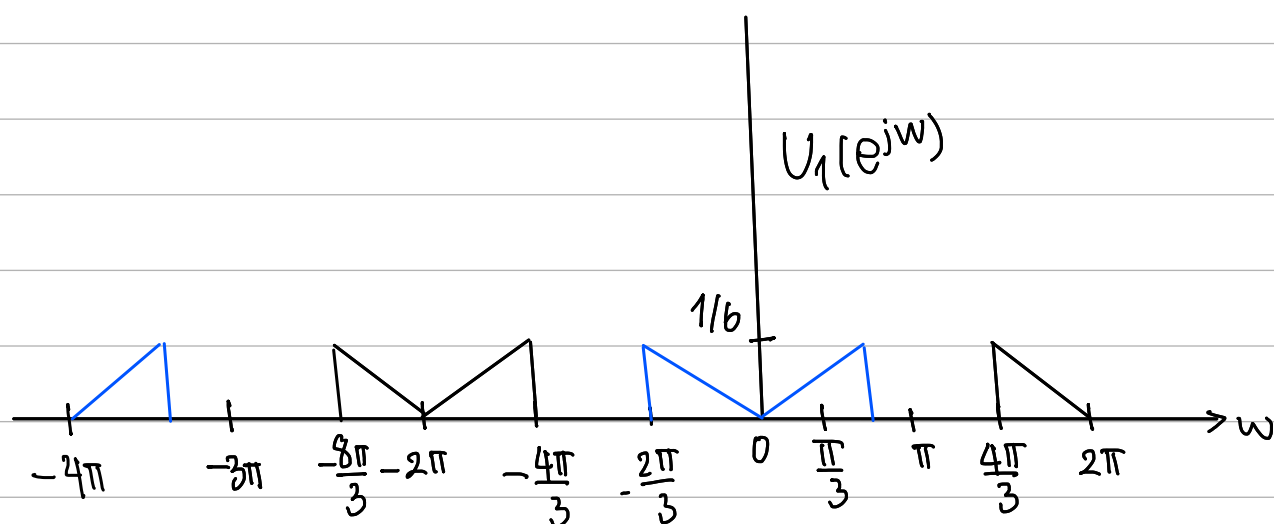
d. This is a channelizer system. It splits the lowpass and highpass spectrum of a signal into different subbands.

3,
a)



$$V_0(e^{j\omega}) = \frac{1}{2}V_0(e^{j\omega}) + \frac{1}{2}V_0(e^{j(\frac{\omega}{2}-\pi)})$$





b. + $U_0(e^{j\omega})$ in problem 2 does not have aliasing because $V_0(e^{j\omega/2})$ does not overlap $V_0(e^{j(\frac{\omega}{2}-\pi)})$

+ $U_0(e^{j\omega})$ in part a experiences aliasing since $V_0(e^{j\omega/2})$ and $V_0(e^{j(\frac{\omega}{2}+\pi)})$ overlap.